Seyed Ali Tabatabaee, Ph.D.

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Canadian Citizen (Eligible for TN Status) LinkedIn GitHub Google Scholar

Summary

I am a Senior Research Engineer at Huawei Canada, owning the cost-based optimizer for GaussDB. I hold a Ph.D. in CS from UBC, where I researched optimizing decisions when inputs are uncertain but can be refined through costly queries. I specialize in algorithms, optimization, distributed systems, and databases.

Education

Ph.D. in Computer Science, The University of British Columbia, Canada Sep 2021 – Jul 2025
• Thesis: Optimization with Explorable Uncertainty, GPA: 95.00/100.00

M.Sc. in Computer Science, The University of British Columbia, Canada Sep 2019 – Aug 2021
• Thesis: Attacking Transaction Relay in MimbleWimble Blockchains, GPA: 91.17/100.00

B.Sc. in Computer Engineering, Sharif University of Technology, Iran Sep 2014 – Jul 2019
• Thesis: Distributed Unit Disk Covering, GPA: 17.16/20.00

Experience

Senior Research Engineer, Huawei, Canada Sep 2025 – Present
• Engineered selective creation and application of statistics views across overlapping queries in a PostgreSQL-based external optimizer for GaussDB, cutting total Join Order Benchmark execution time by over 50%
• Implemented the multithreaded statistics collector for GaussDB's cardinality estimation framework, which achieved over 10% lower total TPC-H execution time, and co-authored a patent application based on the work
• Architected a join sampling framework for GaussDB, extending published work to more complex join types, now ready for implementation

Graduate Research Assistant (Ph.D.), The University of British Columbia, Canada Sep 2021 – Aug 2025
• Led independent research on optimization with explorable uncertainty, defining novel problem formulations, designing algorithms, and proving their guarantees across three peer-reviewed papers and a Ph.D. thesis
• Designed real-time data acquisition strategies for facility location over bounded-speed moving objects, proving the best possible competitive ratios across several variants and impossibility results in high dimensions
• Initiated the study of perpetual scheduling under explorable uncertainty, designing constant-factor competitive and approximation algorithms in the online and offline settings, respectively

Graduate Teaching Assistant, The University of British Columbia, Canada Sep 2019 – Aug 2025
• Delivered tutorials, office hours, and assignment and exam evaluations across 15 terms and 5 courses covering algorithms and theory of computing

Research Intern, The Australian National University, Australia Apr 2025 – Jun 2025
• Extended piqtree with IQ-TREE's sequence alignment simulator using C++ and Python, and built end-to-end workflows generating alignments from coalescent trees simulated with msprime

Research Intern, The University of Edinburgh, Scotland May 2024 – Jul 2024
• Investigated spectral clustering for graphs with hierarchical clusters, proving an expected eigengap under strongly separated edge probabilities across levels and developing Python simulations to examine eigengap patterns

Research Intern, Kyoto University, Japan Jul 2023 – Aug 2023
• Drove the publication of an $O(kn \log n)$ algorithm to construct frequency difference consensus trees, improving the previous best $O(kn \log^2 n)$, by formalizing its proofs, benchmarking it, and writing the final manuscript

Research Intern, The University of Zurich, Switzerland Jun 2022 – Aug 2022
• Developed a Python agent-based simulation to model how miners' transaction-inclusion and block-acceptance strategies affect Bitcoin consensus and miner payoffs when transaction fees are the sole mining incentive

Research Intern, Aquanow, Canada Sep 2020 – Aug 2021
• Led the implementation of a C++ network simulator and a 100-node Azure test network to evaluate attacks on Beam, a privacy-focused blockchain based on MimbleWimble

Graduate Research Assistant (M.Sc.), The University of British Columbia, Canada Sep 2019 – Aug 2021
• Designed and evaluated three attacks on transaction relay in MimbleWimble blockchains, including a new denial-of-service attack that blocks over 45% of transactions with 10% malicious nodes

• Refined the Proof-of-Wait algorithm for Cumulus, a BFT-based sidechain, implemented its on-chain election in Solidity, and evaluated it with JavaScript, showing 93% checkpoint success with 50% silent validators

Undergraduate Research Assistant, Sharif University of Technology, Iran Sep 2017 – Aug 2019

• Designed massively parallel clustering algorithms with constant-factor approximation guarantees, strongly sublinear per-node memory, optimal communication complexity, and complementary lower bounds

Undergraduate Teaching Assistant, Sharif University of Technology, Iran Sep 2016 – Dec 2018

• Managed tutorials and designed and evaluated assignments and quizzes across 9 teaching appointments and 4 courses covering algorithms, theory of computing, and artificial intelligence

Research Intern, Aalto University, Finland Jun 2018 – Aug 2018

• Built SOFIE's decentralized marketplace from scratch, implementing Solidity smart contracts for end-to-end auction workflows, deploying them to a local Ethereum blockchain, and testing them with JavaScript

Publications

• Jesper Jansson, Wing-Kin Sung, **Seyed Ali Tabatabaee**, and Yutong Yang. **A Faster Algorithm for Constructing the Frequency Difference Consensus Tree**. *Journal of Computer and System Sciences*, 2026.

• William Evans and **Seyed Ali Tabatabaee**. **Perpetual Scheduling with Explorable Uncertainty**. In *Proceedings of the 14th International Conference on Algorithms and Complexity*, 2025.

• William Evans and **Seyed Ali Tabatabaee**. **Minimizing the Size of the Uncertainty Regions for Centers of Moving Entities**. In *Proceedings of the 16th Latin American Theoretical Informatics Symposium*, 2024.

• Jesper Jansson, Wing-Kin Sung, **Seyed Ali Tabatabaee**, and Yutong Yang. **A Faster Algorithm for Constructing the Frequency Difference Consensus Tree**. In *Proceedings of the 41st International Symposium on Theoretical Aspects of Computer Science*, 2024.

• Arash Beikmohammadi, William Evans, and **Seyed Ali Tabatabaee**. **Fractional Bamboo Trimming and Distributed Windows Scheduling**. In *Proceedings of the 49th International Conference on Current Trends in Theory and Practice of Computer Science*, 2024.

• Fangyu Gai, Jianyu Niu, Mohammad Jalalzai, **Seyed Ali Tabatabaee**, and Chen Feng. **A Secure Sidechain for Decentralized Trading in Internet of Things**. *IEEE Internet of Things Journal*, 2023.

• **Seyed Ali Tabatabaee**, Charlene Nicer, Ivan Beschastnikh, and Chen Feng. **One Bad Apple Spoils the Bunch: Transaction DoS in MimbleWimble Blockchains**. In *Proceedings of the IEEE 4th International Conference on Blockchain and Cryptocurrency*, 2022.

• Fangyu Gai, Jianyu Niu, **Seyed Ali Tabatabaee**, Chen Feng, and Mohammad Jalalzai. **Cumulus: A Secure BFT-based Sidechain for Off-chain Scaling**. In *Proceedings of the IEEE/ACM 29th International Symposium on Quality of Service*, 2021.

• Kian Mirjalali, **Seyed Ali Tabatabaee**, and Hamid Zarrabi-Zadeh. **Distributed Unit Clustering**. In *Proceedings of the 31st Canadian Conference on Computational Geometry*, 2019.

Honors and Awards

President's Academic Excellence Initiative Ph.D. Award from **UBC** Sep 2021 – Aug 2025

Faculty of Science Ph.D. Tuition Award from **UBC** Aug 2021 – Aug 2025

International Tuition Award from **UBC** Sep 2019 – Aug 2023

Mitacs Accelerate Award (3 Units) May 2020 – Aug 2021

Special UBC Graduate Scholarship - Blockchain@UBC Graduate Student Award Dec 2019 – Aug 2021

Grant for Undergraduate Studies from **Iran's National Elites Foundation** Sep 2014 – Jul 2019

Gold Medal in the **Iranian National Olympiad in Informatics (INOI)** Sep 2013

Bronze Medal in the **Iranian National Olympiad in Informatics (INOI)** Sep 2012

Skills

Technical Skills: C++, C, Python, SQL, PostgreSQL, JavaScript, Java, Solidity, PyTorch, R, Git, Bash, Linux, LaTeX, Azure, AWS, Docker, Django, React, Node.js

Areas of Expertise: Algorithms, Optimization, Distributed Systems, Databases, Software Engineering, Blockchains, Bioinformatics, Game Theory, Machine Learning